Mohamed Oubih

Mechanical Engineer

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Based in Bratislava, Slovakia · Italian (EU) citizen · Open to relocation

4 YEARS	AMS-02	20+	50+
HARDWARE DEVELOPMENT	ISS DETECTOR PROGRAMME	LAUNCH CAMPAIGNS	PRODUCTION DRAWINGS

SUMMARY

Mechanical engineer who takes hardware from requirements through CAD, prototyping, supplier release, and environmental testing. Designed alignment and bonding tooling for the AMS-02 Layer 0 silicon-tracker upgrade, produced 50+ GD&T production drawings in automotive, and is currently the sole mechanical engineer on an ultralight radiosonde and superpressure balloon system. Strongest in lightweight sealing, low-temperature mechanisms, and developing test systems to validate them.

EXPERIENCE

Mechanical & Aerospace Engineer

August 2025 – Present · 1 yr 1 mo

Noove s.r.o. · Bratislava, Slovakia

- **Sole mechanical engineer on Project Picoballoon**, owning all mechanical hardware for the FD-11 radiosonde (~20 g) and the superpressure balloon system (26 µm TritaX envelope): design, drawings, GD&T, BOMs, supplier coordination and release, assembly, and field-test support across 20+ launch campaigns.
- **Developed and characterized a miniature passive pressure-relief valve** through 3 iterations, replacing a direct-acting concept with a **diaphragm architecture** that sets the pressure-sensing area independently of the seat bore, raising design seat contact stress from ~0.04 MPa to ~5 MPa (>100×) at the same cracking pressure; achieved a tunable **1.4–6.5 kPa** range, linear preload response across six configurations (**R^2 = 0.9996**), and derived the flight **venting requirement** from mission telemetry (helium mass flow at 220 K, 17 kPa absolute).
- **Identified two independent low-temperature failure modes:** Silicone **O-ring static seal** leakage at **−30 ± 3 °C**, and a **33% rise in cracking pressure** traced to reduced effective diaphragm area rather than additive stiction (cold/ambient ratio constant at **1.327 ± 0.002** across a 2.6× preload range) — redirecting the redesign from seat to diaphragm.
- Built the **Python/PyQt5 fluidic test platform** used for all valve characterisation: closed-loop differential-pressure control against ambient drift through a networked pressure calibrator, Bronkhorst mass-flow measurement, automated staircase and endurance profiles, and logging and analysis of datasets containing 20k+ samples.
- Owned **FD-11 radiosonde enclosure** development across 7 CATIA/SLA iterations to manufacturing release with outsourced SLA (JLC3DP), covering drawings, GD&T, BOMs, defect-rate analysis and corrective redesign, mass budget, PCB/battery integration, and assembly repeatability.
- Designed **assembly and calibration hardware** for flexible humidity-sensor boards, including SLA fixtures and 0.2 mm laser-cut positioning masks, to improve sensor placement repeatability and reduce operator dependence.

Mechanical Design Engineer

May 2023 – July 2025 · 2 yrs 3 mos

OPmobility · Bratislava, Slovakia

- **Modelled injection-moulded automotive exterior assemblies in CATIA V5** (bumper systems, trim interfaces, brackets, attachment features), applying DFM rules for draft, wall thickness, ribs, snap-fits, tooling direction, parting strategy, undercuts, and sink-mark risk.
- Created and maintained **50+ production drawings** with GD&T, datum schemes, tolerance analysis, and material/process notes to ANSI/ASME standards; managed CAD maturity across multiple vehicle programmes through clearance checks, interface reviews, and assembly-feasibility studies.
- Supported prototype, dimensional, and validation activities for bumper systems on Škoda, Opel, and Citroën platforms, implementing CAD corrections from tooling, quality, manufacturing, and test feedback.

EXPERIENCE · CONTINUED

Research Fellow
August 2022 – May 2023 · 10 mos

University of Perugia · Terni, Italy

- Designed **mechanical integration tooling for the AMS-02 Layer 0 silicon tracker upgrade**, an ISS-based particle-physics detector programme, focused on alignment and adhesive bonding of fragile silicon-sensor subassemblies onto carbon-composite support structures.
- Developed precision alignment fixtures using pin-hole/slot datum interfaces and support buttons, controlling subassembly position within approximately **100 µm across a 1100 × 1200 mm** integration fixture.
- Performed tolerance-stack and datum analysis across tooling and support interfaces, reducing worst-case alignment error from **~120 µm to ~93 µm** through tighter fits and machining specification.
- **Optimised the composite support strategy using ANSYS FEM/FEA**, increasing the support-button layout from 9 to 13 points and reducing predicted self-weight deformation from **~62 µm to ~26 µm** within the deformation budget.
- Built and validated a reduced proof-of-concept prototype, and conducted **PEEK / 3M 2216 epoxy shear testing** including surface preparation, controlled curing, bond-line measurement, and failure-mode analysis.

SKILLS

CAD / Mechanical Design — CATIA V5, SolidWorks, Onshape, solid and surface design, complex assemblies, Geometric Dimensioning and Tolerancing, tolerance analysis, 2D/3D production drawings

Testing / Validation — Pressure, leak, and burst testing; low-temperature and cold-chamber testing to −30 °C; vibration (random, sine, shock); thermal-vacuum chamber; strain gauges; thermocouples; mass-flow and leak-rate measurement

Analysis / Simulation — ANSYS FEM/FEA (stress, deformation, material selection), FMEA, DFM (injection molding, SLA)

Prototyping / Manufacturing — SLA (Formlabs Form 4), FDM, outsourced SLA (JLC3DP), CNC and machined-part sourcing, laser-cut fixtures, assembly tooling, manufacturing-release documentation

Automation / Data — Python, PyQt5, MATLAB, pump / flowmeter / pressure-sensor / valve integration, automated test sequencing and data analysis

Materials / Processes — PTFE, PEEK, Kapton/polyimide, elastomers, SLA resins, carbon-fibre honeycomb, aluminium, adhesive bonding

LANGUAGES

Italian (native) · English (fluent) · Arabic (intermediate) · Slovak (beginner)

EDUCATION

MSc Mechanical Engineering
2020 – 2022 · 2 yrs

University of Perugia · *Cum Laude*

Thesis: design, characterisation, and testing of the alignment and gluing system for integration of silicon cosmic-ray sensors into the AMS-02 Layer 0 tracker upgrade.

BSc Mechanical Engineering
2016 – 2020 · 4 yrs

University of Perugia

Thesis: Measurement of the magnetic characteristics of Manganese-Zinc ferrite cores